

National University of Singapore

(07/08 Semester I Examination)

CZ5101 – Numerical Recipes

27 Nov 2007 – Time Allowed: 2 hours

Instructions to Candidates:

1. This is a closed book examination.
2. This examination paper contains FIVE (5) questions and comprises FOUR (4) printed pages.
3. Marks are indicated at the end of each question; the total marks are 100.
4. Answer ALL the questions.
5. Calculators are allowed.

1. Answer briefly the following questions:

- a. Let x, y, z be some arbitrary IEEE double precision floating-point numbers. Discuss the possible values of the floating-point number $f = [x+(y+z)] - [(x+y)+z]$ that it can take on a computer.
- b. A discrete Fourier transform of length $N = 10007$ is given to some FFT program, e.g., fftw3. How fast can it compute in terms of computational complexity? Note that N is a prime number.
- c. Show that the Lagrange polynomials used in an N -point interpolation satisfy the following condition for any x ,

$$1 = \sum_{i=1}^N l_i(x), \text{ where } l_i(x_j) = \delta_{ij} \text{ are the polynomials of order } N-1.$$
- d. Show that minimizing the 2-norm squared, $\|Ax-b\|^2$, gives the normal equation.
- e. Consider a three-point parabolic integration formula known as the Simpson's rule,

$$\int_{x_1}^{x_3} f(x)dx = (\alpha_1 f_1 + \alpha_2 f_2 + \alpha_3 f_3)h + O(h^n).$$
Derive the coefficient α_i , and give the value of the exponent n .
- f. If the errors in an iterative method of finding a root of an equation decrease like 0.1, 0.001, 0.000000001, ..., how would you characterize the convergence rate?

[30 marks (5 marks each subquestions)]

- a. Assuming that the intermediate values like $(y+z)$ and $x+(y+z)$ are stored also in IEEE format, then the value f is generally not zero, but is about $\max(z,y,z) \times \epsilon$. The machine epsilon $\epsilon \approx 10^{-15}$ for double precision. f can also be a NaN if the computation overflows.
- b. Since N is a prime number, it cannot be factored, the usual trick of FFT can not apply, so we have to compute the slow way, i.e., $O(N^2)$.
- c. Let $f(x) = 1$ be the function to be interpolated with N points, clearly, the values of $y_i \equiv 1$. Use the uniqueness of the polynomial interpolation, we see that equation is true.
- d. $f = \|Ax-b\|^2 = (Ax-b)^T(Ax-b) = x^T A^T Ax - x^T A^T b - b^T A x + b^T b$. Compute the gradient of f , and set it to zero we get $2 \nabla f = A^T Ax - A^T b = 0$, which is the normal equation.
- e. To fix α_i , we consider $1, x, x^2$ to be integrated exactly, to give $1/3, 4/3, 1/3$, $n = 5$.

f. $\varepsilon_{n+1} = \varepsilon_n^3$, the convergence is cubic.

2. Consider a numerical solution to the pair of coupled nonlinear equations,

$$x^2 - y = 0,$$

$$y + xy = 1.$$

- Design a Newton iteration method for the solution.
- Using a calculator, starting from $x_0 = 1$, and $y_0 = 0.5$, iterate the equation(s) and find the solution accurate to two decimal places.

[20 marks, 10 marks each]

- Instead of using two variables, we can eliminate y to get $f(x) = x^3 + x^2 - 1$, the Newton iteration is $x_{n+1} = x_n - f(x_n)/f'(x_n)$.
- Use the above, we get the iteration sequence (1, 0.8, 0.756818, 0.754881, 0.754878, ...), then $y = x^2 = 0.56984$.

3. Consider the problem of minimization of a function.

- Show that in a steepest descent minimization of a function f , the current search direction $\mathbf{n}$ and the next search direction $\mathbf{n}'$ are perpendicular to each other, that is $\mathbf{n}^T \cdot \mathbf{n}' = 0$, where $\mathbf{n}$ and $\mathbf{n}'$ are the negative gradient directions.
- Demonstrate that this is indeed the case for the function $f(x,y) = x + 2x^2 + xy + y^2$, with a starting point of $(x,y) = (1,1)$.
- In conjugate gradient method, the search directions are conjugate, not orthogonal. What does it mean in general mathematically, and what does it mean for the function $f(x,y)$ in part b?
- Use the conjugate gradient method to find the minimum of the above function, starting from the same point $(x,y) = (1,1)$.

[20 marks, 5 marks each]

- Let $\mathbf{n} = -\nabla f(P_0)$, then $\mathbf{n}' = -\nabla f(P_0 + \lambda \mathbf{n})$, we choose λ such that $df/d\lambda = 0$, or $\mathbf{n} \cdot \nabla f(P_0 + \lambda \mathbf{n}) = \mathbf{n} \cdot \mathbf{n}' = 0$.
- $\mathbf{n} = -(1 + 4x + y, x + 2y) = (-6, -3)$, $f(1 - 6\lambda, 1 - 3\lambda) = 5 - 45\lambda + 99\lambda^2$, we find $\lambda = 5/22$. So the next stop is at $(x,y) = (-4/11, 7/22)$. The next negative gradient is at $(3/22, -3/11)$. $\mathbf{n} \cdot \mathbf{n}' = 0$.
- $\mathbf{n} \cdot \mathbf{A} \cdot \mathbf{n}' = 0$, for this problem $\mathbf{n} = (-6, -3)$ and

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$$

n.A.n'=0 means $9n_x + 4n_y = 0$,

d. we can take $n' = (1, -9/4)$. $F(-4/11 + \lambda, 7/22 - 9/4\lambda) = -5/44 + 3/4\lambda + 77/16\lambda^2$, $f'(\lambda) = 0$ gives $\lambda = 6/77$, with result of $x = -2/7$, $y = 1/7$ as the minimum.

4. Consider the problem of linear least squares (or chi-square) fit.
- What is the quantity known as χ^2 that should be minimized with respect to the parameters a_k ?
 - Explain the meaning of “goodness-of-fit” Q . What values of Q are considered a good fit?
 - Let $A\mathbf{a} = \mathbf{b}$, where A is a matrix of $N \times M$, the equality is in the least-square sense. Let $A = U\Sigma V^T$ be the singular value decomposition for A . Express the vector $\mathbf{a}$ in terms of the singular values w_i and the columns or rows of U and V .
 - Computationally, what do you need to do in the formula for the solution in c in order to be numerically stable?

[15 marks, 4,4,4,3 marks]

- $$\chi^2 = \sum_i \frac{(y_i - \sum_k^M a_k X_k(x_i))^2}{\sigma_i^2},$$
- Q is the probability that χ^2 is equal or bigger than the observed value.
Good fit if $0.01 < Q < 0.99$.
- $\mathbf{a} = \Sigma_j (U^T(j).b)V(j)/w_j$
- omit terms j with small w_j .

5. Consider the problem of solution of ordinary differential equations.
- State what it means that an algorithm for a Hamiltonian system is symplectic.
 - A simple one-dimensional harmonic oscillator has a Hamiltonian given by $H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$, where p is momentum and q is coordinate, and ω is the angular frequency of the oscillator. Give an argument (without much derivation if you can) that the following algorithm for a numerical solution of the Hamiltonian equations is symplectic.

$$q_{n+1} = \text{Re} \left\{ \left(q_n + \frac{p_n}{i\omega} \right) e^{i\omega h} \right\},$$

$$p_{n+1} = \text{Re} \left\{ \left(q_n + \frac{p_n}{i\omega} \right) i\omega e^{i\omega h} \right\}.$$
 where $i = \sqrt{-1}$ and h is step size in time.
 - Suppose you used the 4-th order Runge-Kutta method to solve the above oscillator problem instead. What value of h is a reasonable choice? What values of h can we use if we take the method in b? What happen if you run the Runge-Kutta method for an extremely long time on computer?

[15 marks, 5 marks each]

- $\sum_j dp_j \wedge dq_j$ is invariant.
- Since it is the solution of the problem.
- (1) $h = 0.1 \ 2\pi/\omega$, (2) any value for h , (3) diverge to 0 or ∞ (for p or q).

– the end –